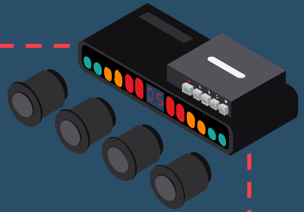
Impact of Air-to-Fuel Ratio in Turbo Engine Performance

Isaac Yong



Topic Origins

Vehicular Efficiency

Wikipedia

Wikipedia

16-25%

- Greenhouse Gases
- Stricter Gas Regulations
- Global Need for Engine Optimization
- Turbochargers/Fuel Mixtures

Definitions

Air-to-Fuel Ratio (AFR)

Gas Turbocharged Engine (GTE)

Brake Specific Fuel Consumption (BSFC)

Knock Tendency

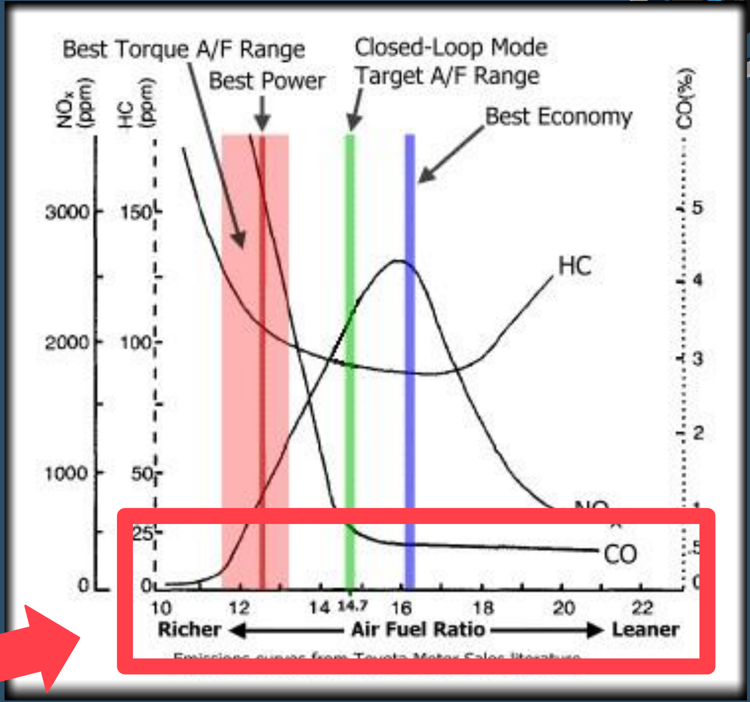
Torque Output

Research Question

How do different AFR values impact GTE performance and what are optimal AFR ranges for GTE operation?

Non-Turbocharged Literature Trends

- AFR Control Strategies
- Differing Fuel Blends
- Non-Turbocharged Engines



(Motor Vehicle, 2019)

Non-Turbocharged Literature Review Trends

Lean (More Air)	Rich (More Fuel)
↑↑Fuel Efficiency↑↑	↓↓Fuel Efficiency↓↓
↑↑Knock Tendency↑↑	↓↓Knock Tendency↓↓
↓↓Torque Output↓↓	↑↑Torque Output↑↑

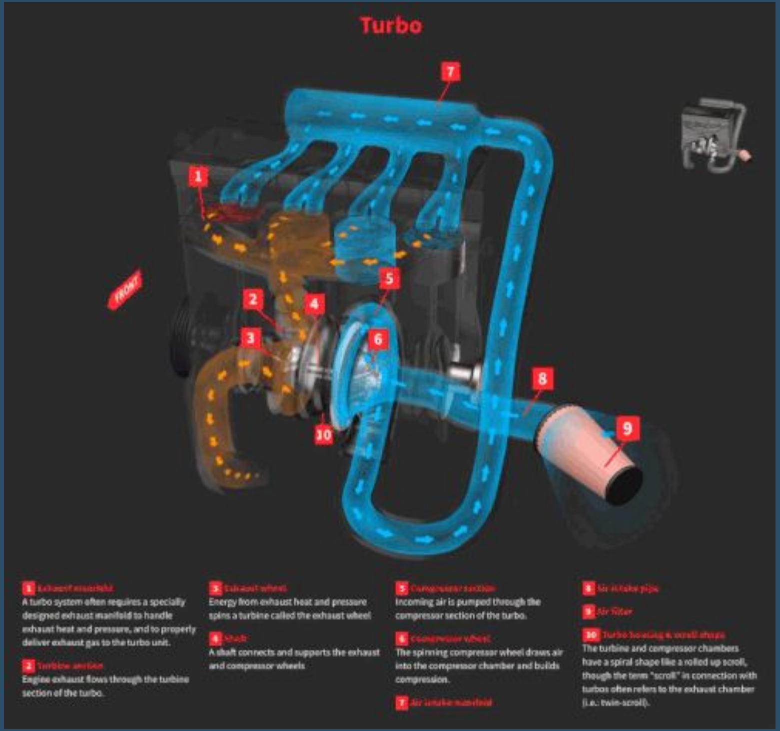
Turbocharged Literature

Amann et al.

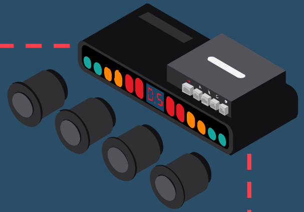
- **Fuel Blends and Aromantics**
- Rich AFR = Less Knock

Andersson et al.

- **Cylinder Air Charge Estimation**
- High Torque Requires Richer AFR (lower AFR)
- Richer AFR = Less Knock



Wikipedia

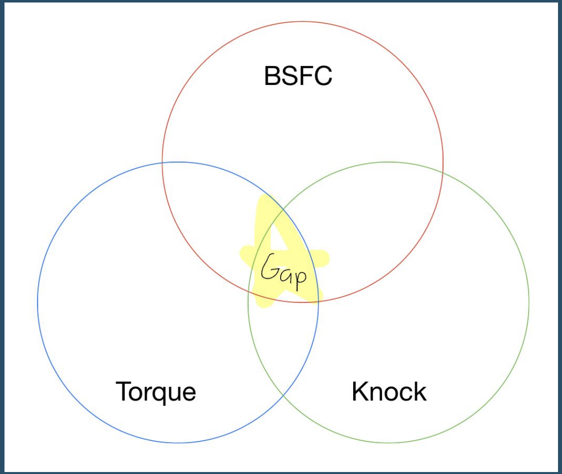


Gap/Goal

Lack of Turbo Studies

Oleg Krivelov

Lack of Variable Breadth



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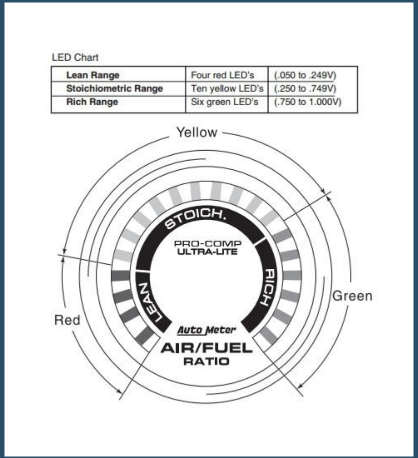
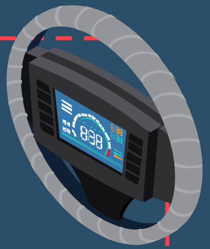
Secondary Focus



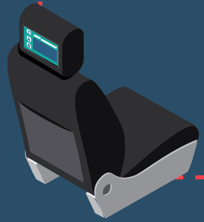
American Muscle



Hypothesis

- Gas Engine Trends ≈ GTE Trends
- Optimal Range:

$$\lambda \approx 0.95\text{-}1$$



Non-Turbocharged Literature Review Trends

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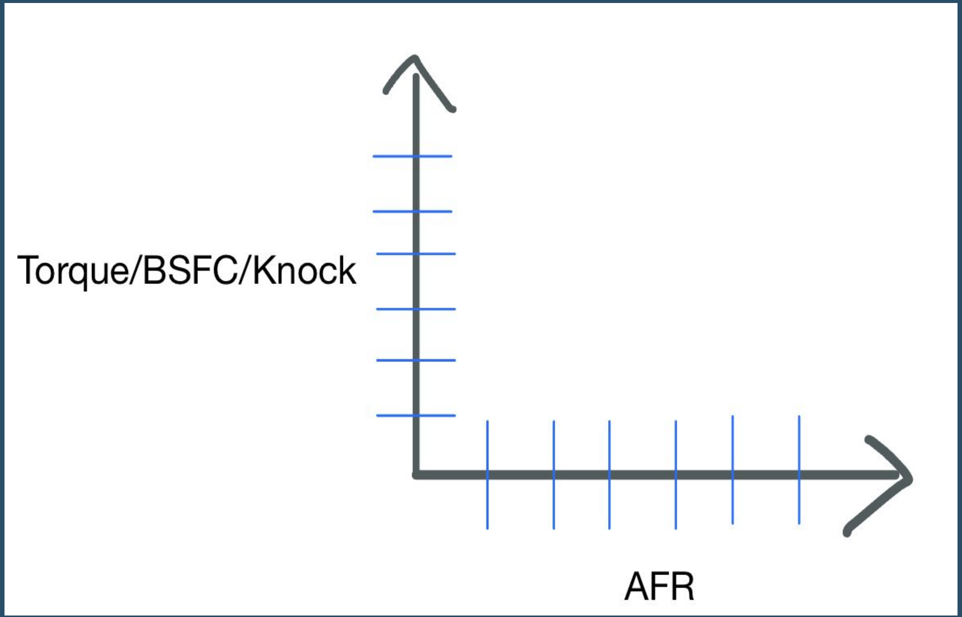
↑↑Fuel Efficiency↑↑	↓↓Fuel Efficiency↓↓
↑↑Knock Tendency↑↑	↓↓Knock Tendency↓↓
↓↓Torque Output↓↓	↑↑Torque Output↑↑

Methodology Description

Quantitative Experimental Design

- Simplified Mean Value Engine Model (MVEM)
- Empirical/Mathematical



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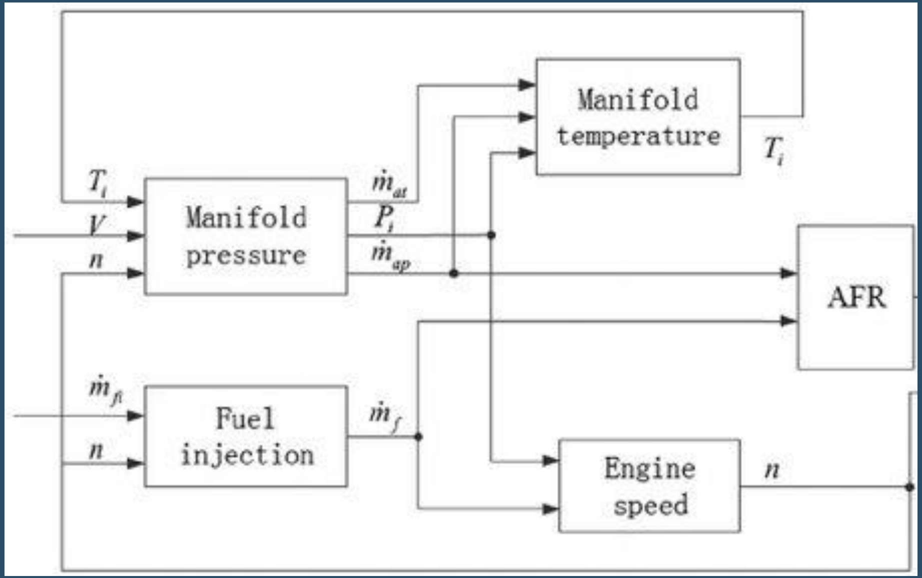
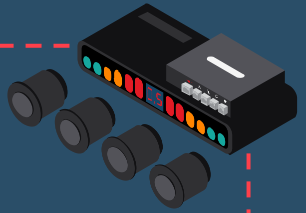
Methodology Justification

Simplified MVEM

- Limited Resources
- Cost-Effectiveness
- Superior Variable Control
- Accuracy



(Yu et al., 2015)



Variable/Parameter Justification

Lamda	2750 RPM	3000 RPM	3500 RPM	4000 RPM	4500 RPM	5000 RPM	5500 RPM	SUM
0.8	1.069897591	0.4229407691	0.1621543822	0.5932924555	0.4904462248	0.591566216	0.2920179478	3.622315586
0.85	1.049927997	0.4592619908	0.2654552756	0.6182202257	0.5274274811	0.6119823886	0.3831382694	3.915413628
0.9	1.050282574	0.5221074288	0.3221632348	0.6757333458	0.5875101499	0.6664342875	0.4306915695	4.254922591
0.95	1.021931336	0.51766046	0.3003759169	0.6647909382	0.5823694823	0.6536039341	0.4021074555	4.142839523
1	0.9269781057	0.4855349106	0.2781155925	0.6271302749	0.5497050343	0.6150300135	0.3751262359	3.857620167
1.05	0.8185316113	0.4300461806	0.2307646784	0.5669238745	0.4937972596	0.5547353574	0.3247555121	3.419554474
1.1	0.698692769	0.3544657119	0.1618166699	0.4873374752	0.4178964369	0.475772721	0.2542560776	2.850237862
1.15	0.5689780758	0.261281261	0.08244993956	0.3907808666	0.324478976	0.3804664524	0.1746453415	2.183080913
1.2	0.4304593242	0.1632476755	-0.01507579385	0.2907022517	0.2256954858	0.2821953289	0.07804469643	1.455268969

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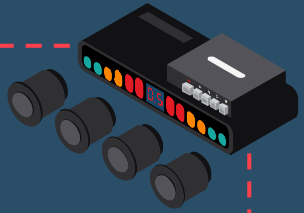
- Engine Safety
- Power Output
- Fuel Economy
- Wide Open Throttle Conditions
- AFR increment = 0.05
- AFR Range: 0.8-1.2

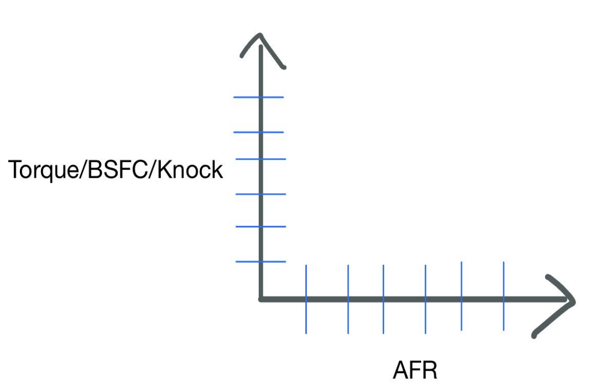
Parameter Justification



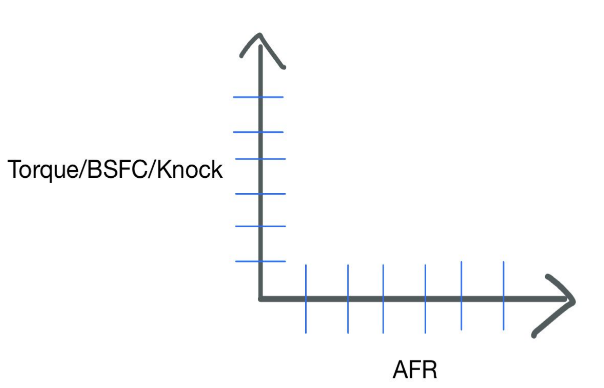
- Revolutions Per Minute (RPM)

  2.75k to 5.5k

- Generally Increments of 500
- Wide Range of Driving Conditions
- Trend Consistency
- Input into Model



RPM 3k



RPM 3.5k

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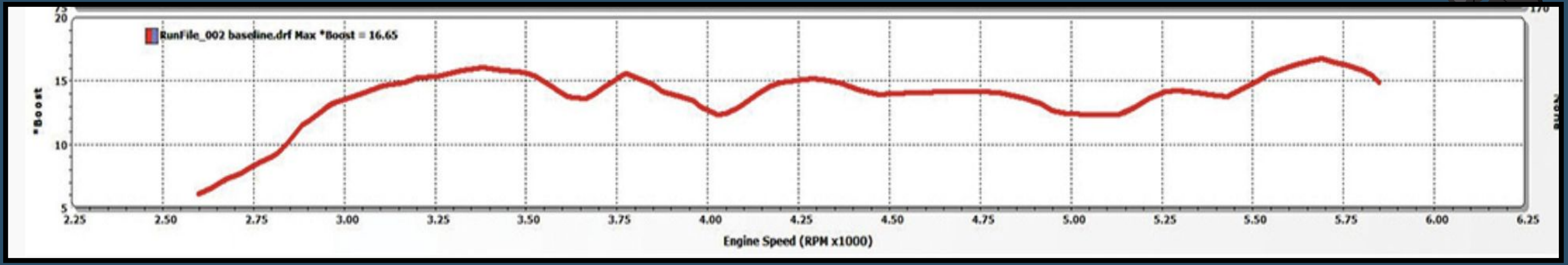
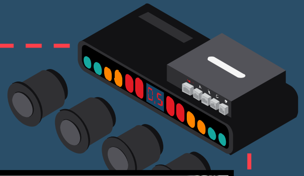
Mathematical Model



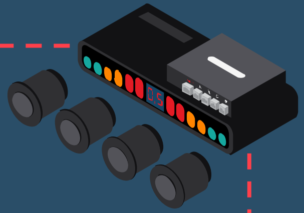
(DSPORT Magazine, 2017)

$$p_r = \frac{p_{man}}{p_{amb}} = \frac{p_{boost} + p_{amb}}{p_{amb}}$$

2017 Subaru WRX STI Turbo Engine (EJ257) Dynamometer Log



Mathematical Model

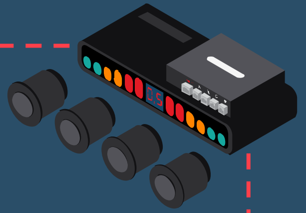


AFR Equations

$$m_{air} = \frac{[\eta_{vol}(N, p_{man})](V_d)(N)(p_{man})}{120\,(R)(T_{man})}$$

$$\lambda = \frac{m_{air}}{14.7\,(m_{fuel})} = \frac{AFR_{actual}}{AFR_{stoichiometric}}$$

Mathematical Model



Knock Equations

$$T_{ID} = 17.69\left(\frac{ON}{100}\right)^{3.402} * p_{peak}^{-1.7} * e^{3800/T_{peak}}$$

$$T_{RES} = \frac{(13.5 * 1000)}{6N}$$

$$\text{(SHR Eq.) } p_{peak} = p_{man} * (CR)^{\gamma}$$

$$\text{(SHR Eq.) } T_{peak} = T_{man} * \left(\frac{p_{peak}}{p_{man}}\right)^{[1-\frac{1}{\gamma}]}$$

$$KM = T_{ID} - T_{RES}$$

Torque/BSFC Equations

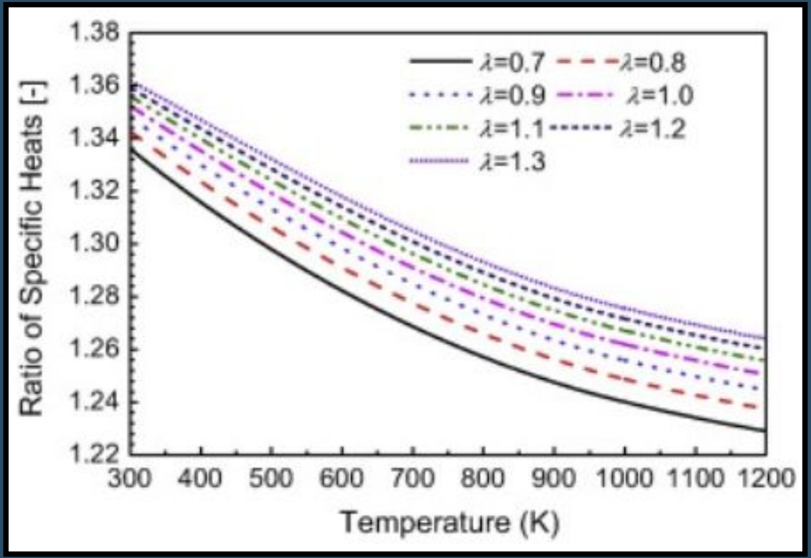
$$\tau = \frac{9549\,(BP)}{(N)}$$

$$BSFC = \frac{(m_{fuel})}{BP}$$

$$BP = (\eta_{th})\,(m_{fuel})(Q_{heat})$$

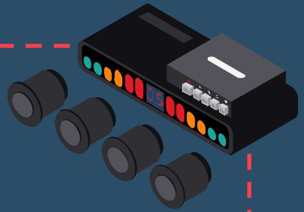
Mathematical Model

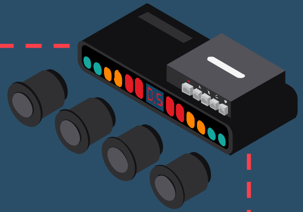
Knock Estimation



(Chen, 2014)

$$T_{EGT} = \frac{T_{peak} + T_{man}}{2}$$





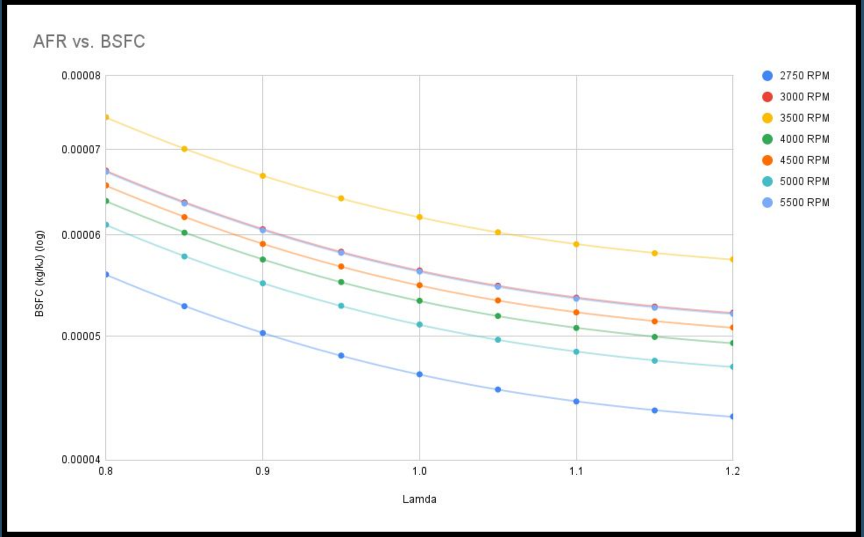
Data Analysis

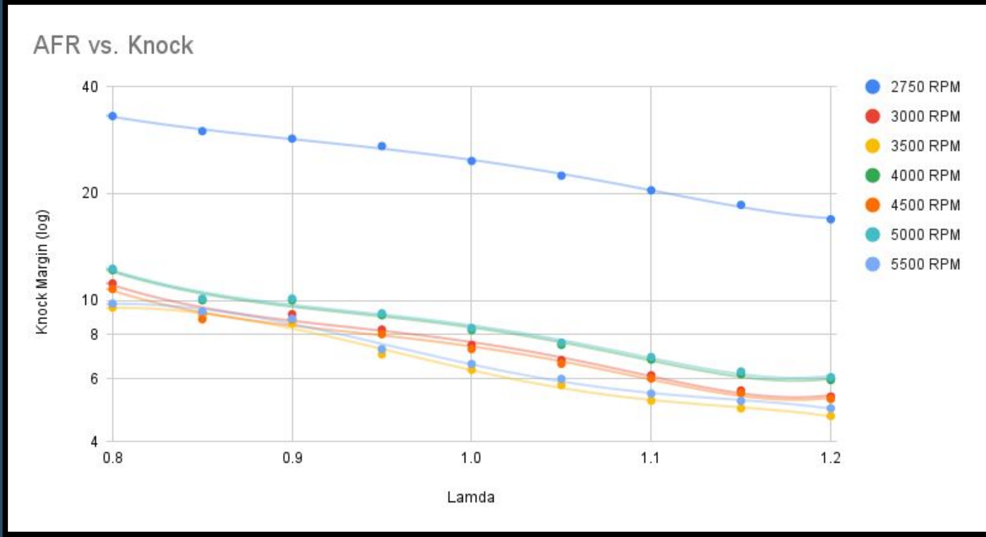
- Min-Max Normalization
- Equal Weights to variables
- Normalized Score Value

$$\text{Normalized Score } (Z) = \frac{value - min}{max - min}$$

$$Z_{perf} = Z_{\tau} + Z_{knock} - Z_{BSFC}$$

Results and Implications





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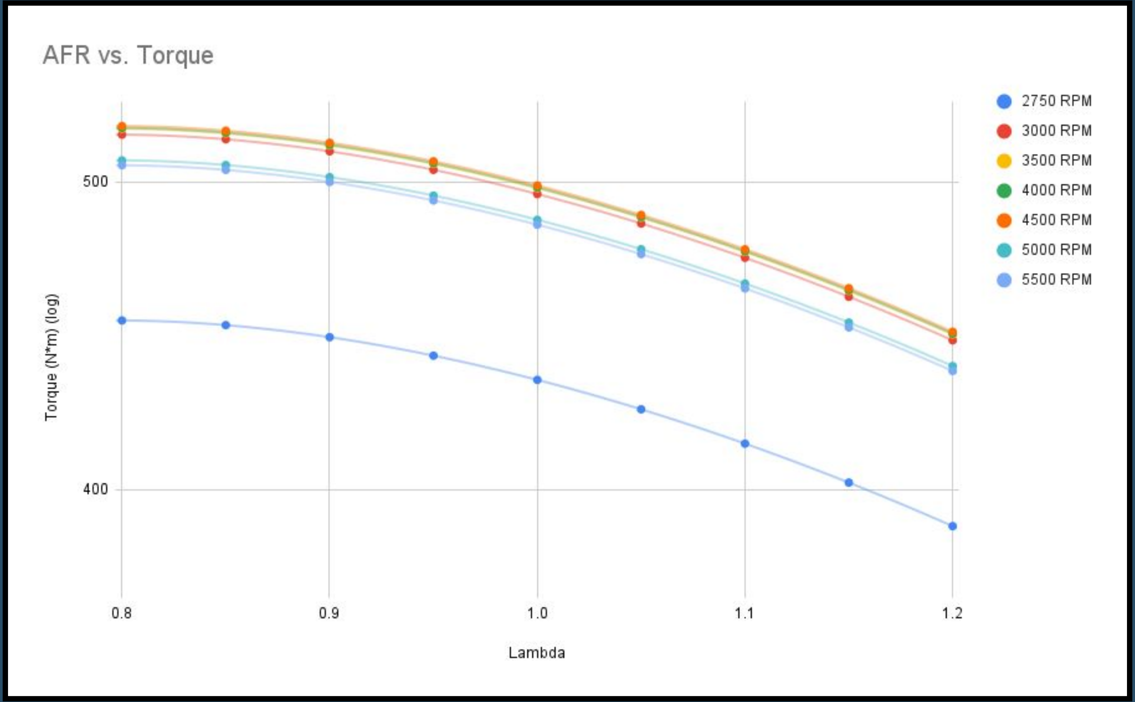

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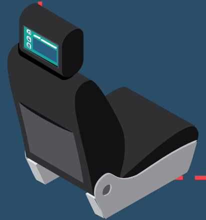
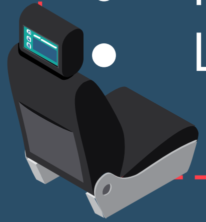




Implications

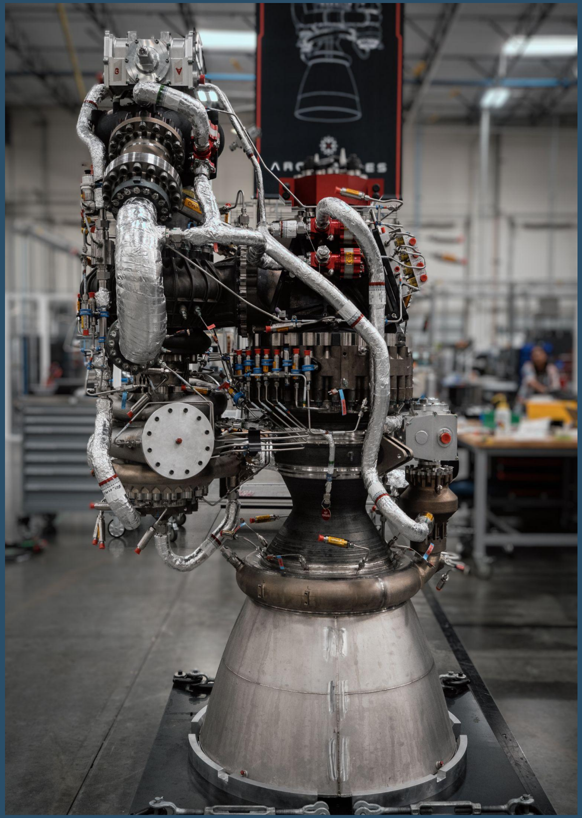
- Similar Gas and GTE Trends
- All-Encompassing/Simple Optimization Calculator
- Wide Variety of Vehicles/Machines
    - Gas/Diesel/Hybrid
    - Turbine Machines
- Reduce Global Warming
- Long term Sustainability

Limitations

- Theoretical in Nature
- Isentropic/Ideal Conditions
- Limited Data Variability
- Gap in Scholarly Data

Future Research

- Focus on Emissions
- Other Engine Types
- Different Engine Set Ups
- Real Engine Set Up



Business Wire

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